PiNNTORP LLC

Online Gambling Platform
Phase 1: Project Planning Submission

CSCI 2040U - Software Design and Analysis

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1. Project Overview

Project Name	PiNNTORP Online Gambling Platform
Team Name	PiNNTORP LLC
Course	CSCI 2040U - Software Design and Analysis
Members	Adrian Ahmadi, Mele Felix, Hayden Dunn, Eric Beaulne, Nikola Grujin

Project Purpose & Scope

PiNNTORP LLC aims to build an online gambling web app designed to give players a proper and entertaining gambling experience. Users will be able to play games like slots and a few other games, connect with their friends, track their own wins, and get recommendations based on what they and their friends enjoy playing over time.

High-Level Goals & Expectations

- Deliver a functional web-based gambling platform with around 3 fully fleshed out gambling games
- Implement a recommendation algorithm for users to get recommended games to play based on their user activity and friend activity
- Provide a friends list system that allows users to see who is online and what they are playing
- Display limited user statistics to the user, such as wins, and keep all the other statistics hidden to be used in the algorithms and other features
- Allow users to delete their accounts and all the associated data

2. Team Agreements & Elicitation Documentation

Team Contract Summary

Roles & Responsibilities

Role	Member
Project Manager	Eric Beaulne
Technical Manager	Mele Felix
Front-End Lead	Adrian Ahmadi
Back-End Lead	Nikola Grujin
Software Quality Lead	Hayden Dunn
Developers	All members

Key Expectations & Conflict Resolution

- Communication via Discord with a 48-hour response expectation.
- Weekly in-person meetings during scheduled lab sessions.
- Pair programming for all coding outside of lab meetings.
- Mandatory code reviews by a non-contributing member before merging to main.
- Decisions made as a team in consultation with the customer.
- If falling behind, ask for help rather than miss deadlines.

Development Pipeline

Aspect	Details
Languages / Frameworks	Java, HTML, CSS, JavaScript, Docker
Repository / VCS	GitHub
Planning Board	Trello (Kanban)
Database	SQL-based

Elicitation Methods Used

- Domain Research: Researched AML/KYC compliance requirements, responsible gambling standards, RNG certification, and other similar platforms such as BetMGM, Bet365, Rainbet, etc.
- Stakeholder Identification: Identified AGCO and iGaming Ontario as the regulatory stakeholders
- Source Analysis: Reviewed external sources on legal gambling in Ontario, the terminology used in the domain, complying with AML, and the regulated markets
- Customer Interview: Met with the customer during this phase to clarify the project requirements, the feature scope, and the priorities.

Key Elicitation Findings

- The customer envisions having a standard gambling mechanism (slots, real-time), with personalization for the users through the recommendation algorithms.
- The customer desires a friend system that includes the addition, reading, updating, and deletion of the friends of the user
- There are no gambling mechanics out of the scope, the customer aims to have around 3 functional games
- Animations are not required, functional correctness is the priority
- There are no legal or age-restriction requirements for this project

Customer Meeting Notes

The following questions were asked to the customer during the preliminary meeting to clarify the project's overall scope:

Q1: Which gambling features are explicitly required (e.g., betting, loot boxes, spins), and are there any mechanics that are not allowed or out of scope?

No restricted mechanics, everything is allowed

Q2: Do you expect fair randomness (e.g., seeded RNG, disclosed odds), and should users be able to view probabilities or odds?

Not discussed

Q3: Are there specific legal, ethical, or age-restriction requirements we must follow for this project, even if the app is only a simulation?

There are no specific requirements for this specific project

Q4: What aspects matter most for evaluation: correctness, UI/UX, security, realism, documentation, or extensibility?

Correctness and documentation are the most important, as having the functionality be accurate is important for the platform that we are building, and documentation is essential to be evaluated.

Q5: What is the minimum viable feature set?

At least 3 functional games, a database, and a friends list.

Q6: How polished should animations be versus functional correctness?

Animations not required explicitly, games just need to work properly.

Q7: Should the app track betting history, win/loss ratios, or session length?

It should track the user's history, their wins, losses, games, etc. It should just display the wins to the user to not discourage them from proceeding.

3. User Stories & Prioritization

User stories categorized into Must Have, Should Have, or Nice to Have based on customer priorities.

User Story	Priority	Effort	Dependencies
As a user, I want functional game mechanics so I can use the product.	Must Have	14 days	None (foundational)
As a user, I want to see my statistics to make decisions.	Must Have	4 days	Game Mechanics
As a user, I want my win-to-loss ratio so I can brag.	Must Have	4 days	Game Mechanics, Stats
As a user, I want game recommendations so I don't get bored.	Should Have	10 days	Game Mechanics, Stats
As a user, I want to add friends to see who is active.	Should Have	4 days	None
As a user, I want to delete my account for data security.	Should Have	6 days	Account system
As a user, I want pretty aesthetics and flashing win colours.	Nice to Have	6 days	Game Mechanics

Prioritization Justification

- Must Have - Game Mechanics (Priority 50): The platform depends on having at least one functional game
- Must Have - Statistics (Priority 30): Essential for having good user engagement, as its the core customer expectation
- Should Have - Algorithm (Priority 40): Key to the personalization vision but not blocking the core.
- Should Have - Friends List (Priority 20): Adds value and feeds recommendations but not required.
- Should Have - Account Deletion (Priority 10): This is important for user privacy, but it has a lower initial urgency
- Nice to Have - Game Visuals (Priority 20): Customer confirmed that the animations not required therefore they are nice to have

4. Effort Estimation & Iteration Planning

Planning Poker Results

Planning Poker was conducted based on the features discussed and the general scope, and cards were selected (1, 2, 4, 6, 10, 14). The discrepancies were discussed, and a decision was reached.

Feature	Disagreed & Discussed	Consensus
Friends List	no	4 days
Algorithm	yes	10 days
Account Deletion	no	6 days
Game Mechanics	yes	14 days
Game Visuals	yes	6 days
Statistic Display	no	4 days

Task Assignments

Feature	Lead	Support	Iter.
Game Mechanics (logic, RNG)	Nikola Grujin	Mele Felix	1
Game Mechanics (front-end)	Adrian Ahmadi	Hayden Dunn	1
Statistic Display	Mele Felix	Nikola Grujin	2
Recommendation Algorithm	Nikola Grujin	Adrian Ahmadi	2
Friends List	Adrian Ahmadi	Hayden Dunn	3
Account Deletion	Mele Felix	Eric Beaulne	3
Game Visuals	Adrian Ahmadi	All	3*
Project Coordination	Eric Beaulne (PM)	All	1-3
QA & Code Review	Hayden Dunn	All	1-3

Iteration Plan (Three Two-Week Iterations)

Iteration 1: Foundation + Core Gameplay (Weeks 1–2)

Focus: Build game engine and deliver at least one playable gambling game.

Task	Assigned To	Duration	Priority
Game logic, RNG	Nikola Grujin, Mele Felix	10 days	50
Game front-end, UI	Adrian Ahmadi, Hayden Dunn	4 days	50
Project setup	Eric Beaulne, All	2 days	50

Iteration 2: Data + Intelligence (Weeks 3–4)

Focus: Statistics tracking and recommendation algorithm.

Task	Assigned To	Duration	Priority
Statistics & W/L tracking	Mele Felix, Nikola Grujin	4 days	30
Recommendation algorithm	Nikola Grujin, Adrian Ahmadi	10 days	40

Iteration 3: User Control + Compliance (Weeks 5–6)

Focus: Social features, account management, visual polish if time permits.

Task	Assigned To	Duration	Priority
Friends list	Adrian Ahmadi, Hayden Dunn	4 days	20
Account deletion	Mele Felix, Eric Beaulne	6 days	10
Game visuals (if time)	Adrian Ahmadi, All	6 days	20
Final QA & testing	Hayden Dunn, All	4 days	-

Agile Board (Trello - Kanban)

PiNNTORP LLC. uses Trello as the Kanban board for the project, the screenshot below shows the initial state. Following are planned board states per iteration.

Current Trello Board (Initial State)

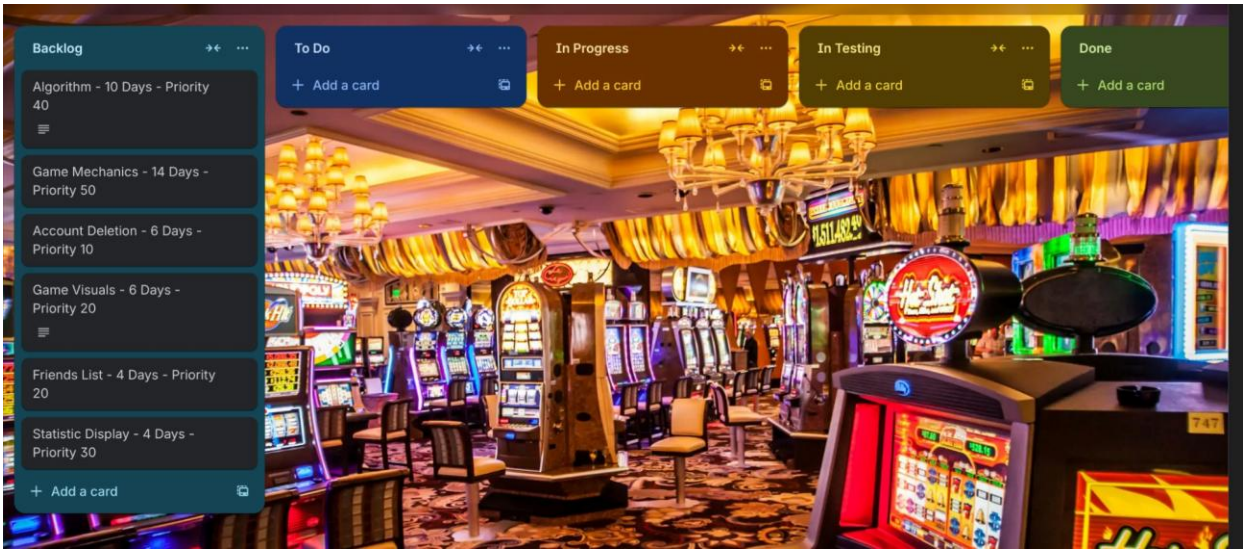

Figure 1: Trello board - initial backlog state with all feature cards and priorities.

Planned Board: Iteration 1 (Weeks 1–2)

Backlog	To Do	In Progress	In Testing	Done
Algorithm - 10 Days - P40	Game front-end - 4 Days - P50	Game Mechanics - 14 Days - P50		
Account Deletion - 6 Days - P10		Project Setup – Repo, etc		

Game Visuals - 6 Days - P20				
Friends List - 4 Days - P20				
Statistic Display - 4 Days - P30				

Planned Board: Iteration 2 (Weeks 3–4)

Backlog	To Do	In Progress	In Testing	Done
Account Deletion - 6 Days - P10	Algorithm - 10 Days - P40	Statistic Display - 4 Days - P30		Game Mechanics - 14 Days - P50
Game Visuals - 6 Days - P20				
Friends List - 4 Days - P20				

Planned Board: Iteration 3 (Weeks 5–6)

Backlog	To Do	In Progress	In Testing	Done
Game Visuals - 6 Days - P20		Friends List - 4 Days - P20	Algorithm - 10 Days - P40	Game Mechanics - 14 Days - P50
		Account Deletion - 6 Days - P10		Statistic Display - 4 Days - P30

5. Finalized Project Roadmap

Major Milestones & Deliverables

Deadline	Milestone	Deliverables
End of Week 2	Iteration 1	Playable game with correct mechanics; infrastructure operational.
End of Week 4	Iteration 2	Statistics dashboard live; recommendation algorithm functional.
End of Week 6	Iteration 3 & Final	Friends list, account deletion working; integration tests passed.

Key Risks & Potential Challenges

Risk	Impact	Mitigation
The game logic complexity exceeds the estimate	High - delays its dependents	Start with a simple game like slots, then expand after core is stable.
The accuracy of received recommendations	Medium - poor suggestions	Simple skeleton first, then flesh it out fully once more information is given.
Our team availability	Medium - uneven workload	Pair programming + Discord check-ins.
Integration issues	Medium - boundary bugs	Code reviews + integration tests each iteration.
Scope creep from aesthetics	Low - diverts effort	Animations confirmed not required by customer, it is nice to have, but not essential.

Next Steps & Immediate Actions

- Repository setup: Initialize GitHub repo with more project structure, database schema, etc.
- Trello board: Move Iteration 1 cards from Backlog to To Do and assign owners.
- Game selection: Finalize which game to implement first.
- Pair programming kickoff: Begin Iteration 1 with assigned pairs.

Appendix: Glossary of Terms

Term	Definition
Bet (or Wager)	Risking money on the outcome of some event.
High Roller	A Gambler who consistently wagers large sums of money.
Book	Short for sportsbook or bookmaker; person or establishment that takes bets from customers.
Parlay	A parlay combines 2 or more bets into one. For the parlay to be paid out, each bet needs to be successful. Because of the risk, parlays are more lucrative than simple bets.
Favourite / Underdog	The expected winner or loser, respectively, of a game or event. This is factored into a payout.
Hedging	Making a bet opposite to your original bet. This is done to ensure profit or to minimize potential loss
Payout %	Winnings paid to players divided by the bet handle
Gambler's Fallacy	The belief that, if an event has occurred less frequently than expected, it is more likely to happen again in the future (or vice versa).
Bet Handle	Total amount wagered for a bet.
Point Spread	(Sports betting) How many points the favourite is favoured over the underdog
Cover	The result of a point spread bet. For a favourite to cover, they have to win by more than the point spread. For an underdog to cover, they have to win outright or lose by less than the spread.
Rake	A fee taken by the house from each poker pot or tournament entry as revenue.
Hit Frequency	Probability that a slot game produces a win on a given spin.
House Advantage	Advantage built into casino machines and games, so that on average, the house (casino) always gains money
Proposition	A more specific bet in sports betting, such as how many points a certain player will score